



Laser cut Mild Steel

.030"-.500"

12 Thicknesses

2-4 days

Production Time

GET INSTANT PRICING
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ABOUT USES AVAILABLE SERVICES SPECS RESOURCES

1008 and A36/A1018 steels for your next project

(<https://sendcutsend.com>)

MATERIALS

Reference the chart below to learn more about our Cold Rolled and HRP&O offerings. For further information about each steel, reference the individual material pages linked in the table.

1008 Cold Rolled Steel

7 thicknesses: .030" – .135"

(<https://sendcutsend.com/materials/mild-steel/cold-rolled/>)

A36/A1018 HRP&O

5 thicknesses: .187" – .500"

(<https://sendcutsend.com/materials/mild-steel/hrpo/>)

What can you make with mild steel sheet metal?

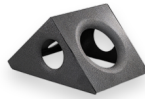
If you need something tough, affordable, and easy to fabricate, mild steel gets it done. It cuts, bends, and welds clean. Perfect for brackets, frames, and anything that needs to hold up without driving up cost.

Services available for Mild Steel

(<https://sendcutsend.com>)



MATERIALS ▾

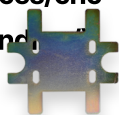


Bending

Add dimension to your projects with our precision CNC bending services.



(<https://sendcutsend.com/services/cnc-bending>)



Deburring

Remove small imperfections, scratches, and burr left over from the cutting process.



(<https://sendcutsend.com/services/deburring>)



Dimple Forming

Strengthen material with aesthetically pleasing dimples.



(<https://sendcutsend.com/services/dimple-forming>)



Hardware Insertion

Select from our catalog of PEM press-fit hardware to add nuts, studs, and standoffs.



(<https://sendcutsend.com/services/hardware>)



Plating

Increase rust prevention, wear resistance, and strength with zinc and nickel plating.



(<https://sendcutsend.com/services/plating>)

Powder Coating

Give your custom cut parts a bold, long-lasting protective layer in one of 15 options.



(<https://sendcutsend.com/services/powder-coating>)

Welding

Weld the seams on your bent parts closed for a stronger, finished edge.



(<https://sendcutsend.com/services/welding>)

Tapping

Easily add threading to allow for the addition of hardware to your parts.



(<https://sendcutsend.com/services/tapping>)

Services/plating/

(<https://sendcutsend.com/>)

Services/powder

-coating/)

Services/welding

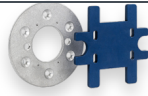
/)

Services/tapping

/)



MATERIALS ▾



Tumbling

Reduce the surface blemishes and handling scratches found in raw materials.



(<https://sendcutsend.com/services/tumbling/>)

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Mild Steel sheet metal material details and specifications

(https://sendcutsend.com)

MATERIALS



InchMM

.030".048".059".074".104".119".135".187".250".313".375".500"

.030" A1008 Mild Steel

✓ Laser Cutting

VIEW SPECS

✓ Bending

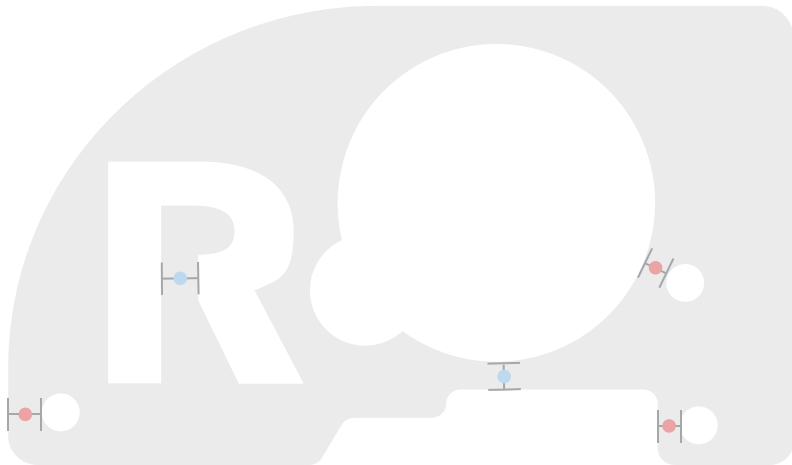
VIEW SPECS

✓ Dimple Forming

VIEW SPECS

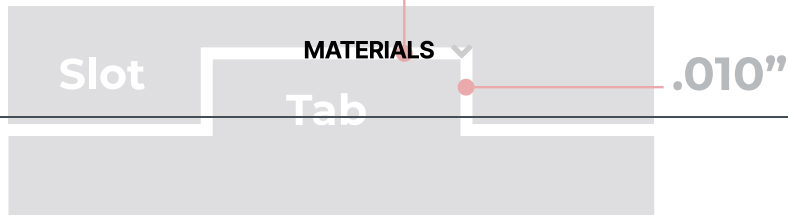
Material Details & Specifications

A1008 Mild Steel 0.030" GENERAL DETAILS	
Advertised Thickness	0.030"
Gauge	22
Thickness tolerance positive	0.003"
Thickness tolerance negative	0.003"

Mill Finish	(https://sendcutsend.com)	Cold Rolled	Q
Top/Bottom finish	MATERIALS Identical both sides		
Sourced from	USA/Global		
LASER CUTTING SPECIFICATIONS			
Cutting process	Fiber laser		
Cut tolerance +/-	0.005"		
Flatness tolerance before cutting	+/-0.030 per foot		
Min part size	.25" x .375"		
Max part size	44" x 30"		
Min hole size	.015"		
Min bridge size	.015"		
Min hole to edge distance	.020"		
 Hole to Edge Bridge			
Tab and slot tolerance	.010"		

(<https://sendcutsend.com>)

.010"



BENDING SPECIFICATIONS

Min bend part size	.375" x 1.5"
Max bending flat part size	44" x 30"
Max bend length	44"
Min flange length (before bend/Flat Pattern) 90° or less (obtuse)	0.255"
Min flange length (after bend) 90° or less (obtuse)	0.286"
Minimum Length Center of bend line 91-130* (Acute)	0.344"
Die width	0.472"
Effective bend radius @ 90°	0.045"
Max bend angle	130°
Minimum bend angle	5°
Bend angle tolerance (bend length up to 24")	+/- 1 degree
Bend angle tolerance (bend length over 24")	+/- 2 degrees
Bend deduction @ 90°	0.061"
K Factor	0.38

Bend relief depth (https://sendcutsend.com)	0.095"	MATERIALS	
Minimum joggle. Bend line to bend line @90° max flange	0.288"		
Maximum joggle. Bend line to bend line @90° max flange	3.750"		
DIMPLE SPECIFICATIONS			
Min dimple part size	1.200" x 5.200"		
Max dimple part size	26" x 56"		
Smallest dimple	.500"		
Largest dimple	3.000"		
Top View  Side View  Dimple Overall Height			
Dimple overall height (.500" dimple)	.125"		
Dimple overall height (.750" dimple)	.127"		
Dimple overall height (1.000" dimple)	.170"		
Dimple overall height (1.250" dimple)	.183"		
Dimple overall height (1.500" dimple)	.200"		
Dimple overall height (1.750" dimple)	.208"		
Dimple overall height (2.000" dimple)	.225"		

Dimple overall height (2.500" dimple) (https://sendcutsend.com)	.285"	
Dimple overall height (3.000" dimple)	.365"	

MATERIALS

A1008 MILD STEEL PROPERTIES	
Material Composition	Iron (Fe): 99.31 – 99.7 Manganese (Mn): 0.3 – 0.5 Carbon (C): 0 – 0.1 Sulfur (S): 0 – 0.05 Phosphorus (P): 0 – 0.04
Density	490 lb/ft^3
Heat treatments process	N/A
ASTM	A1008-21A-CS-TYPE-B
Tensile Strength (Ultimate)	54 ksi
Tensile Strength (Yield)	45 ksi
Shear Strength	34 ksi
Shear Modulus	11000 ksi
Fatigue Strength	32 ksi
Brinell Hardness	100
Elongation at Break	22%
Elastic Modulus	27000 ksi
Poisson's Ratio	.29
Thermal Conductivity	43 BTU/h-ft °F
Melting Point	2600 °F
Magnetic	Yes



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(https://sendcutsend.com)

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Additional charts and guidelines



Min/Max Part Sizes

→ [\(https://sendcutsend.com/materials/min-max/\)](https://sendcutsend.com/materials/min-max/)



Processing Min/Max Part Sizes

→

(<https://sendcutsend.com/materials/processing-min-max/>)



(<https://sendcutsend.com>)
Design Guidelines



(<https://sendcutsend.com/guidelines/>)

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Bending Deformation Guidelines → (<https://sendcutsend.com/guidelines/bend-deformation/>)



Hardware Catalog → (<https://sendcutsend.com/guidelines/hardware/catalog/>)



Drill Size Reference Chart → (<https://sendcutsend.com/guidelines/drill-chart/>)



Bending Calculator → (<https://sendcutsend.com/bending-calculator/>)



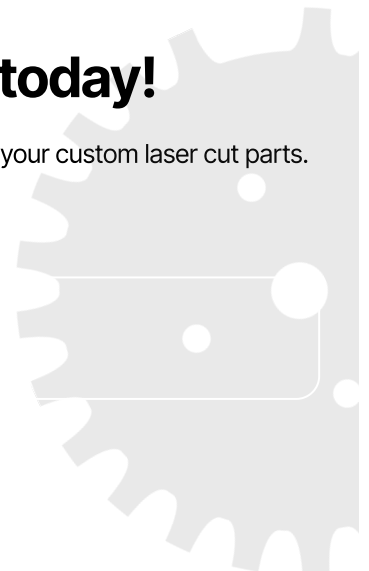
Assembly Guidelines → (<https://sendcutsend.com/guidelines/assembly/>)

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Mild steel sheet metal properties and fabrication guide

Mild steel is the workhorse of sheet metal fabrication — strong, weldable, affordable, and incredibly versatile. At SendCutSend, we offer several common mild steel grades including 1008, A36, and A36/A1018, all laser cut with $\pm.005''$ precision and available with a range of secondary services. Below, we break down the essentials of mild steel, how its grades compare, and how to design your parts for reliable performance and clean fabrication.

Key takeaways for mild steel

- **Versatile and affordable:** A dependable option for brackets, enclosures, panels, and structural parts.
- **Multiple grades available:** **1008** (<https://sendcutsend.com/materials/mild-steel/cold-rolled/>) for formability, **A36** (<https://sendcutsend.com/materials/mild-steel/hot-rolled/>) for strength, and **A36/A1018** (<https://sendcutsend.com/materials/mild-steel/hrpo/>) for balanced performance.
- **Excellent weldability and machinability:** Compatible with MIG, TIG, and spot welding.
- **Laser cut precision:** Clean, burr-free edges with $\pm.005''$ tolerances and fast turnaround.
- **Full range of secondary services:** SendCutSend offers bending, hardware insertion, plating, powder coating, tapping, and tumbling for complete fabrication.

What is mild steel?

Mild steel is a low-carbon steel containing roughly 0.05–0.25% carbon, offering a balance of strength, ductility, and weldability. It's the backbone of the metal fabrication world, used in everything from automotive brackets and fixtures to furniture and machinery. Because of its low carbon content, mild steel is easy to shape, cut, and weld — making it ideal for both **prototyping** (<https://sendcutsend.com/prototyping/>) and **production parts** (<http://sendcutsend.com/commercial/production-runs>).

What are the properties of mild steel?

Mild steel combines strength, formability, and cost-effectiveness. Typical properties (vary by grade):

- **Density:** 0.283 lb/in³ (7.83 g/cm³)
- **Tensile strength:** 45–70 ksi
- **Yield strength:** 25–50 ksi
- **Elongation:** 20–35%

It's strong enough for structural use yet soft enough to bend or tap easily.

What's the difference between 1008, A36, and A36/A1018 mild steel? <https://sendcutsend.com>



Each grade of mild steel serves a slightly different purpose:

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- **Cold Rolled 1008 mild steel** (<https://sendcutsend.com/materials/mild-steel/cold-rolled/>): Very low carbon (~0.08%) for maximum formability and clean bends. Common in sheet applications and parts that will be stamped or dimple formed.
- **Hot Rolled A36 mild steel** (<https://sendcutsend.com/materials/mild-steel/hot-rolled/>): Stronger, with higher carbon (~0.25%) for better structural integrity and weld strength. Excellent for brackets, mounts, and load-bearing parts.
- **HRP&O A36/A1018 mild steel** (<https://sendcutsend.com/materials/mild-steel/hrpo/>): A blended specification with tighter thickness and surface finish tolerances — ideal for precision sheet metal components.

All three are laser cut with the same $\pm 0.005''$ precision and support SendCutSend's bending, tapping, and coating services. If you're not sure which is right for your design, explore each material's dedicated page in our **Material Catalog** (<https://sendcutsend.com/materials/>).

How does mild steel compare to stainless steel or aluminum?

Mild steel offers higher strength than aluminum and better formability than stainless, but requires protective coatings for corrosion resistance.

- **Mild steel Versus aluminum** (<http://sendcutsend.com/materials/laser-cut-aluminum>): Stronger and stiffer, but heavier and less corrosion-resistant.
- **Mild steel Versus stainless steel**: (<http://sendcutsend.com/materials/stainless-steel>) Easier to weld and form, but more prone to rust if left uncoated.
- **Mild steel is bBest for**: General-purpose parts where cost, strength, and machinability outweigh environmental exposure.

For more comparisons, see our **Metal Comparison Guides** (<https://sendcutsend.com/materials/metals>).

How does mild steel perform during laser cutting?

Mild steel laser cuts exceptionally well, producing clean edges and tight tolerances.

- **Edge quality**: Smooth and square with minimal dross.
- **Tolerance**: $\pm 0.005''$ across all thicknesses from .030"—.500".
- **Heat-affected zone**: Minimal, with no warping or distortion under typical conditions.

Proper design for kerf and feature size ensures a perfect fit during assembly — see our **Design Guidelines** (<https://sendcutsend.com/guidelines/>) for details.

Can mild steel be bent or formed? (<https://sendcutsend.com>)



Yes, mild steel bends cleanly and consistently, especially in thinner gauges.

MATERIALS ▼

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- Inside bend radius = 1x material thickness or greater for most applications.
 - 1008 mild steel offers the best ductility for complex bends or dimples.
 - Avoid bending across laser-cut features to maintain structural integrity.

Check our **Bending Guidelines** (<https://sendcutsend.com/guidelines/bending/>) for recommended radii and clearances by thickness.

Can mild steel be welded or joined?

Mild steel is one of the easiest metals to weld.

- Works with MIG, TIG, or stick welding.
- No preheat required in most thicknesses.
- Use ER70S-6 filler for clean, strong welds.

Because it welds so predictably, it's a go-to material for custom brackets, roll cages, and structural assemblies.

What finishing options are available for mild steel?

Since mild steel is prone to oxidation, protective coatings are highly recommended. At SendCutSend, you can choose from several finishing services:

- **Powder coating** (<http://sendcutsend.com/services/powder-coating>): Durable, uniform color protection.
- **Plating** (<http://sendcutsend.com/services/plating>): Adds corrosion resistance and a polished appearance.
- **Tumbling** (<http://sendcutsend.com/services/tumbling>): Smooths sharp edges and surface burrs.
- **Hardware insertion** (<http://sendcutsend.com/services/hardware>) and **tapping** (<http://sendcutsend.com/services/tapping>): Add threads and fasteners directly during fabrication.

Review SendCutSend's **services guidelines** (<http://sendcutsend.com/guidelines>) before adding additional services to your laser cut mild steel parts.

How durable is mild steel over time?

Mild steel is strong and resilient, but it will oxidize (rust) if left unprotected.

- Indoors or painted: Long lifespan with minimal maintenance.
- Outdoors or humid environments: Requires plating or powder coating to prevent rust.
- Structurally, mild steel retains strength for decades when properly protected.

Protective finishes make mild steel a long-term performer for both indoor and outdoor parts.

(<https://sendcutsend.com>)



Design tips for mild steel sheet metal

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Keep strength, bendability, and coating needs in mind during design.

- Add relief cuts near bends for accurate angles.
- Include drainage or vent holes for coated parts.
- Maintain standard hole sizes for tapping or hardware insertion.
- Choose a surface finish early to avoid double handling.

For more fabrication-specific details, review our [Design Guidelines \(https://sendcutsend.com/guidelines/\)](https://sendcutsend.com/guidelines/).

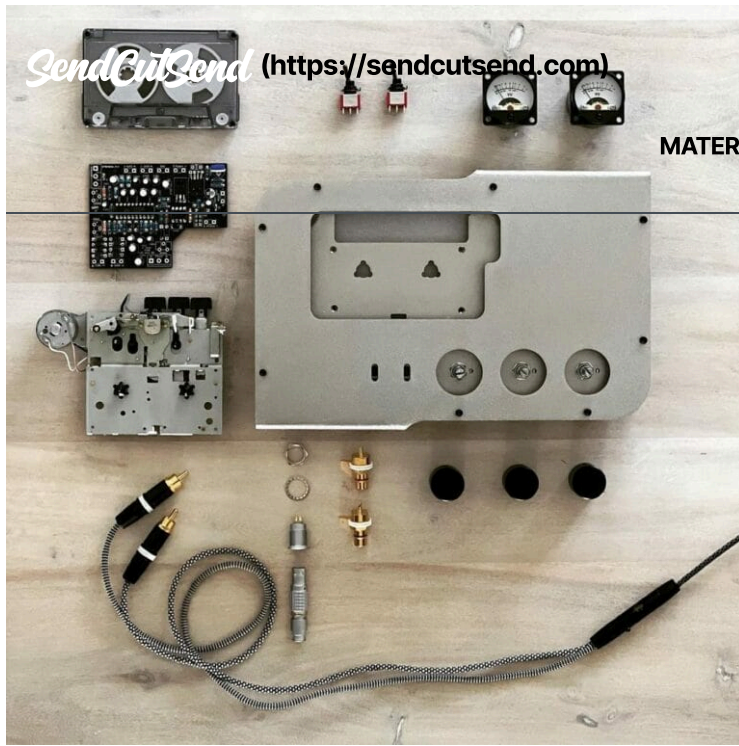
Why choose mild steel?

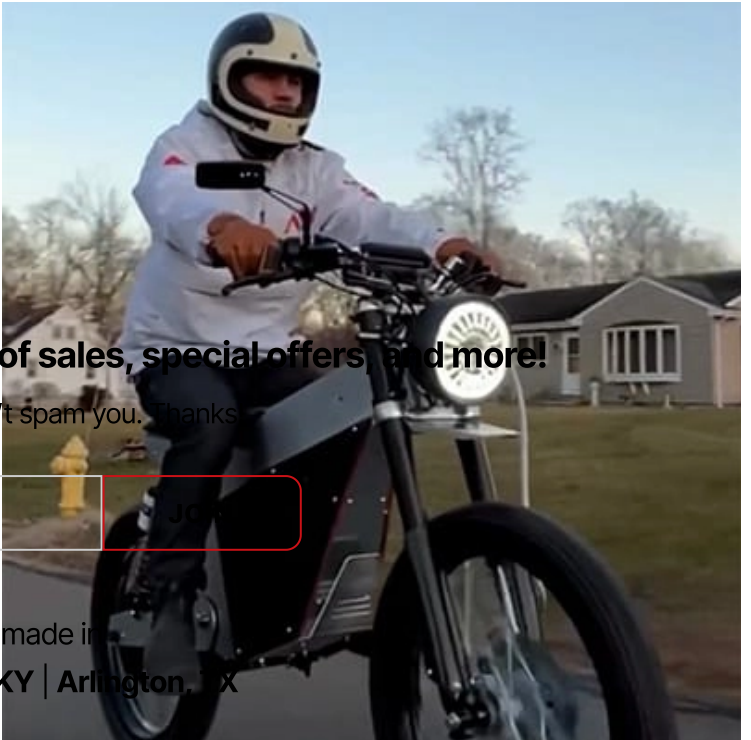
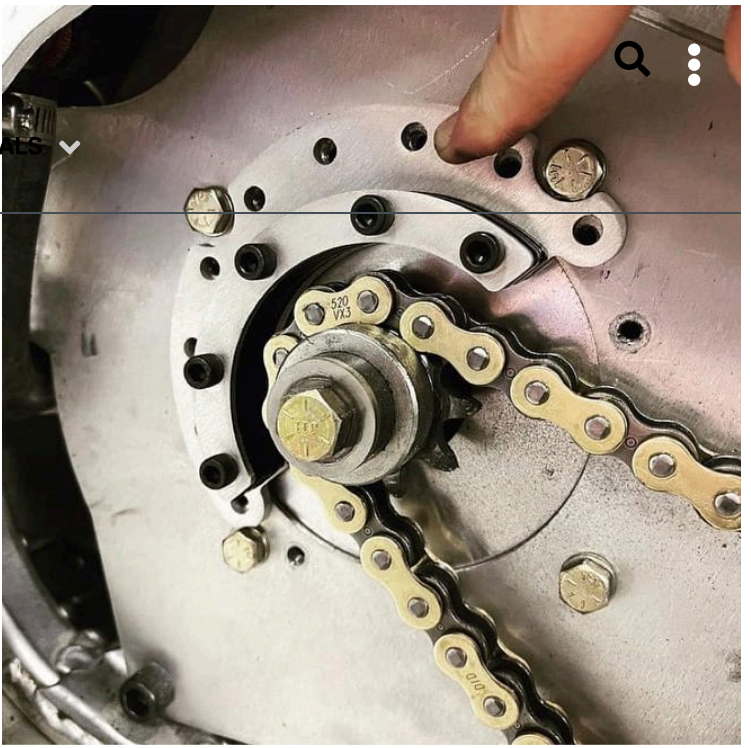
Mild steel is the all-purpose backbone of fabrication — affordable, predictable, and easy to work with. Whether you need strength (A36), formability (1008), or precision (A36/A1018), SendCutSend laser cuts your parts with $\pm.005''$ accuracy and offers bending, coating, and hardware services to deliver a ready-to-assemble product. Explore all three mild steel grades in our [Material Catalog \(https://sendcutsend.com/materials/mild-steel/\)](https://sendcutsend.com/materials/mild-steel/) to find the right fit for your project.

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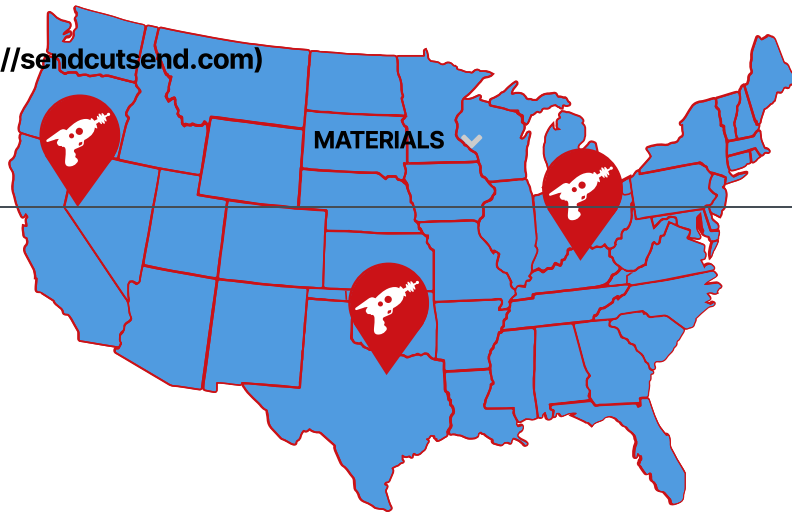
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